

Stage 2 The CAPM

Introduction

Using the same 29 historical Dow Jones stocks as Stage 1, we find a flatter empirical Security Market Line (SML), a positive fitted intercept and three marginal individual alpha signals. The chance-count assessment does not establish a definitive joint rejection of zero alphas.

The sample contains 130 monthly USD returns, November 2015-August 2026, with the Stage 1 return and spin-off conventions unchanged. The risk-free proxy is the Kenneth French one-month Treasury-bill return; the market proxy is its broad US equity market return. Both match the currency and frequency of our stocks, although the equity proxy does not cover all investable wealth.

2a Security characteristic regressions

$$y_{i,t} = R_{i,t} - R_{f,t}, \quad x_t = R_{M,t} - R_{f,t}$$

$$y_{i,t} = \alpha_i + \beta_i x_t + \varepsilon_{i,t}$$

For each stock, OLS estimates the intercept and market exposure from 130 months. Stock returns subtract contemporaneous RF once; Mkt-RF is already excess return. Inputs remain monthly decimals. The OLS coefficients obey:

$$\hat{\beta}_i = \frac{\widehat{\text{Cov}}(x, y_i)}{\widehat{\text{Var}}(x)}, \quad \hat{\alpha}_i = \bar{y}_i - \hat{\beta}_i \bar{x}$$

Table 2.1 CAPM estimates for all 29 stocks

Stock	Beta	Alpha %	t alpha	R sq.	Stock	Beta	Alpha %	t alpha	R sq.
MMM	0.9239	-0.460	-0.928	0.366	JPM	1.0748	0.459	1.041	0.497
AXP	1.1913	0.119	0.249	0.510	MCD	0.4610	0.308	0.775	0.182
AAPL	1.1361	0.838	1.530	0.416	MRK	0.2773	0.765	1.440	0.043
BA	1.3889	-0.680	-0.819	0.317	MSFT	0.9092	0.935	2.015	0.389
CAT	1.2400	0.966	1.459	0.368	NKE	0.9430	-1.143	-1.851	0.279
CVX	0.8682	0.202	0.319	0.238	PFE	0.4951	-0.217	-0.391	0.117
CSCO	0.9328	0.361	0.695	0.349	PG	0.3298	0.307	0.749	0.097
KO	0.4088	0.322	0.842	0.159	TRV	0.5716	0.462	1.004	0.203
XOM	0.7267	0.231	0.352	0.169	RTX	0.8883	0.387	0.675	0.285
GE	1.1667	-0.107	-0.134	0.259	UNH	0.6586	0.504	0.738	0.134
GS	1.3732	0.177	0.360	0.564	VZ	0.2857	0.153	0.321	0.056
HD	0.9592	-0.040	-0.092	0.443	V	0.8254	0.386	1.015	0.438
INTC	1.3697	0.194	0.160	0.174	WMT	0.4577	0.952	2.113	0.146
IBM	0.9314	-0.080	-0.132	0.282	DIS	1.2882	-1.170	-2.068	0.463
JNJ	0.4079	0.482	1.186	0.143					

Alpha is monthly; every regression has 130 observations and uses conventional OLS standard errors. R squared measures time-series fit, not the CAPM pricing restriction. Blue identifies retained DJIA members; amber identifies later index removals. Old DuPont is excluded for the historical data gap; UTX continues as RTX.

2b The empirical Security Market Line

The CAPM links expected excess return to beta. We use arithmetic sample means and estimate a cross-sectional line through the 29 stock observations:

$$\bar{y}_i = \frac{1}{T} \sum_{t=1}^T y_{i,t}, \quad \bar{x} = \frac{1}{T} \sum_{t=1}^T x_t, \quad T = 130$$

$$\bar{y}_i = a + b\hat{\beta}_i + u_i, \quad i = 1, \dots, 29$$

Here a is the cross-sectional intercept and b is the fitted slope, distinct from each stock's time-series alpha and beta. The theoretical SML restrictions in excess-return space are:

$$\text{CAPM: } a = 0, \quad b = \bar{x} = 0.010642$$

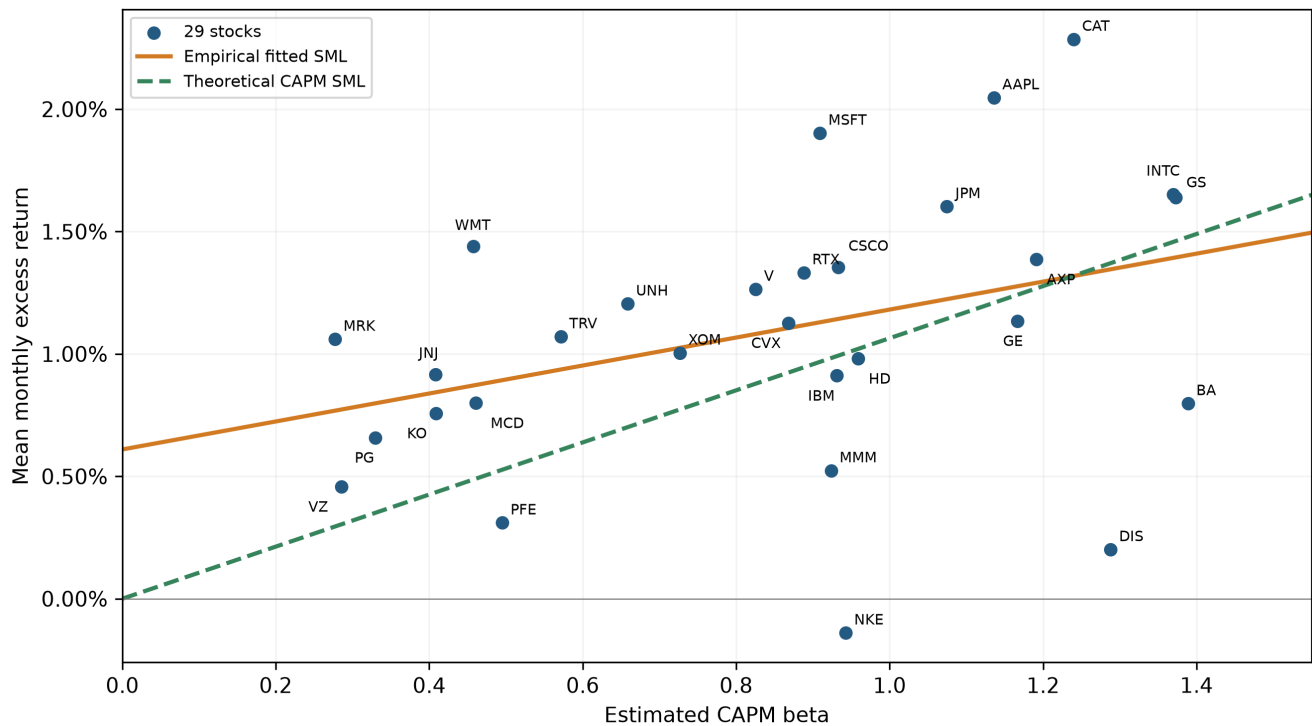


Figure 2.1 The fitted SML is flatter than theory, and individual stock means show substantial dispersion around both lines.

Table 2.2 Fitted and theoretical SML parameters

Line	Intercept % monthly	Slope % monthly
Empirical fitted SML	0.610	0.571
Theoretical CAPM SML	0.000	1.064

The theoretical line passes through the origin because RF has already been subtracted. Its slope is the common-window mean market premium, 1.064% per month. Cross-sectional R squared is 0.133: the fitted beta relationship explains about 13.3% of the variation in stock mean excess returns. This differs from the time-series R squared in Table 2.1.

2c A joint assessment of zero alphas

The CAPM requires all population alphas to be zero. The joint null is:

$$H_0: \alpha_1 = \alpha_2 = \dots = \alpha_{29} = 0$$

We use the handout's permitted chance-count discussion rather than a formal GRS test. Each alpha is compared with its conventional OLS standard error:

$$t(\hat{\alpha}_i) = \frac{\hat{\alpha}_i}{SE(\hat{\alpha}_i)}, \quad |t(\hat{\alpha}_i)| > 2$$

The threshold approximates a two-sided 5% test. With 130 observations and two coefficients, each regression has 128 residual degrees of freedom. The approximate rule and exact $p < 0.05$ criterion identify the same three stocks.

Table 2.3 Individual alphas exceeding the handout threshold

Stock	Alpha % monthly	t alpha	p alpha	Direction
MSFT	0.935	2.015	0.0460	Positive
WMT	0.952	2.113	0.0366	Positive
DIS	-1.170	-2.068	0.0406	Negative

MSFT and WMT have positive sample pricing deviations; DIS has a negative deviation. All three p values are near 5%, indicating marginal individual signals. AAPL's positive alpha of 0.838% is not significant under this rule ($t = 1.530$).

Observed counts versus chance

$$K = \sum_{i=1}^{29} \mathbf{1}\{|t(\hat{\alpha}_i)| > 2\}, \quad K_{\text{observed}} = 3$$

$$E[K | H_0] \approx \sum_{i=1}^{29} 0.05 = 29 \times 0.05 = 1.45$$

The indicator counts one when a stock exceeds the threshold and zero otherwise. If every true alpha is zero and each test has approximately 5% size, linearity of expectation gives about 1.45 chance significant results across 29 tests.

Table 2.4 Observed versus expected significant counts

Quantity	Number
Stocks tested	29
Observed $ t \text{ alpha} > 2$	3
Approximate expected count under the joint null	1.45

Three exceeds 1.45, but the expectation is not an upper limit. The expected count does not require independent tests; the count distribution does depend on their dependence. Shared stock shocks make a simple independent-binomial calculation inappropriate here. We therefore do not infer a formal joint p value from this comparison.

The evidence flags possible deviations for Stage 3 without a definitive joint rejection. Non-significant individual estimates do not prove zero alpha. Conventional OLS p values also depend on the error assumptions and may be sensitive to heteroskedasticity or serial correlation. No Holm adjustment or GRS test is used.

2d The fitted SML versus theory

In decimal monthly-return units, the two lines are:

$$\text{Fitted: } \hat{y}(\beta) = 0.006099 + 0.005713\beta$$

$$\text{Theory: } \bar{y}^{\text{CAPM}}(\beta) = 0.010642\beta$$

Table 2.5 SML parameter differences from theory

Parameter	Fitted % monthly	Theory % monthly	Difference pp
Intercept	0.610	0.000	+0.610
Slope	0.571	1.064	-0.493

A flatter slope

$$\frac{\hat{b}}{\bar{x}} = 0.537, \quad 1 - \frac{\hat{b}}{\bar{x}} = 46.3\%$$

The fitted slope is only 53.7% of theory, or 46.3% lower. A one-unit increase in beta is associated with 0.571 percentage points more monthly mean excess return, compared with the theoretical 1.064 points. Higher market exposure is accompanied by a smaller realized return increment than the sample CAPM benchmark predicts.

$$\beta^* = \frac{\hat{a}}{\bar{x} - \hat{b}} = 1.237$$

The lines cross at beta 1.237. Below this level the fitted line is above theory; above it the fitted line is below theory. At beta 0.5, the fitted mean is 0.896% versus 0.532% under theory. These statements concern the fitted relationship, not every individual stock.

A positive intercept

The fitted intercept exceeds the theoretical zero by 0.610 percentage points per month. It is not RF or an individual stock's alpha. Observed betas range from 0.277 to 1.389, so the intercept at beta zero is an extrapolation, not evidence of a tradable riskless profit opportunity.

$$\hat{\alpha}_i = \bar{y}_i - \hat{\beta}_i \bar{x}$$

Each stock's time-series alpha is its sample mean deviation from the theoretical SML. The cross-sectional intercept and slope instead summarize the pattern across all stocks. This explains why the fitted-line comparison and the individual-alpha assessment complement each other.

Interpretation and limits

Week 4 links nonzero alphas to possible mispricing, missing factors or a poor market proxy. Our flatter SML and positive intercept are descriptive deviations; estimated betas, noisy means and common shocks prevent us from treating the parameter differences as a conclusive test of the population CAPM.

Historical membership retains later removals, mitigating current-member selection, but old DuPont's exclusion leaves availability selection. The equity proxy differs from the unobservable portfolio of all wealth. Stage 3 should assess whether factors absorb the MSFT, WMT and DIS signals; Stage 4 should examine sensitivity to another market proxy.

Sources: FE5108 midterm project handout, Track A Stage 2; Week 4 CAPM slides 9 and 34-35; Stage 1 frozen data and methods. Market and RF: Kenneth French US monthly factors, 202608 vintage. Stocks: instructor bundle and cached Yahoo Finance series under the Stage 1 cleaning conventions.